

3.4 Matrix Multiplication: The Fourth Perspective



Core skills you'll master in this section:

- ▶ The row-vector-linear-combination perspective: for the matrix multiplication AB , each row vector of the result is a weighted sum of the row vectors of the right-hand matrix B .
- ▶ In the matrix multiplication AB , A 's every row serves as a weight applied to B 's every row; the weighted sum generates C 's every row.
- ▶ The row-vector perspective and the column-vector perspective on matrix multiplication convert into each other via transposition.
- ▶ In the matrix multiplication AB , A has however many rows, AB has that many rows.

The fourth perspective on matrix multiplication views matrix multiplication as a linear combination of row vectors. This perspective, starting from the row-vector point of view, reveals another way of looking at matrix multiplication — for the matrix multiplication $C = A @ B$, C 's every row is the result of a linear combination of B 's row vectors.

Matrix A 's every-row elements serve as coefficients that perform a weighted sum on the row vectors of matrix B , generating the corresponding row of the result matrix C .

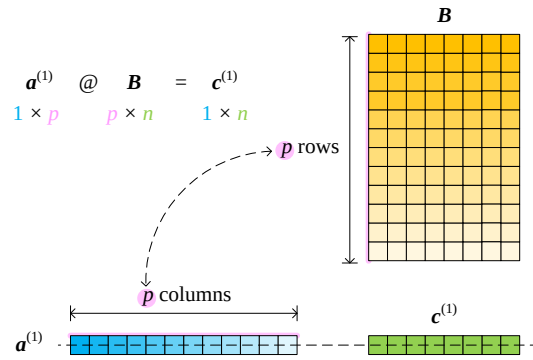
Matrix C's First Row

You might be wondering — since we can view every column of matrix C as a linear combination of A 's column vectors, can we also view matrix C as some kind of linear combination of matrix B ?

The answer is yes!

Take matrix C 's first row as an example; as shown in Figure 1, $c^{(1)}$'s corresponding matrix multiplication is

$$c^{(1)} = a^{(1)} @ B$$

Figure 1. $\mathbf{c}^{(1)} = \mathbf{a}^{(1)} \mathbf{B}$ – the Corresponding Matrix Multiplication

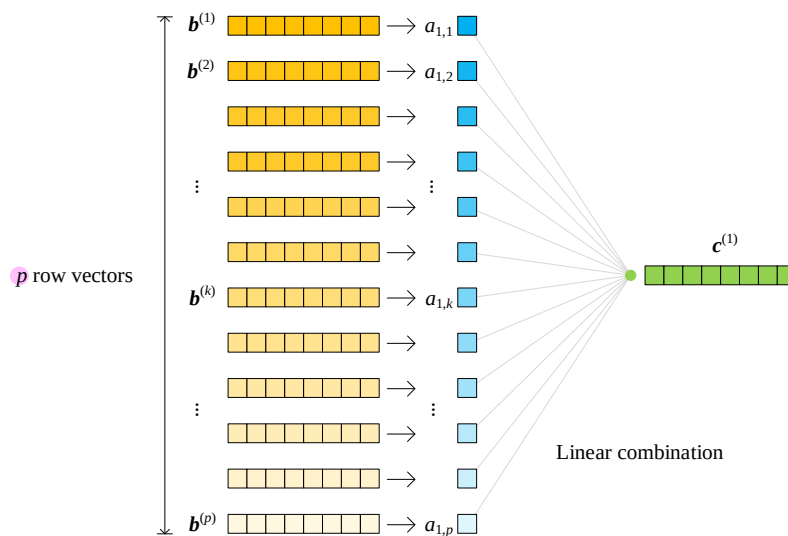
Write matrix $\mathbf{B}$ as a set of row vectors

$$\mathbf{B} = \begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(D)} \end{bmatrix}$$

As shown in Figure 2, $\mathbf{c}^{(1)}$ is a linear combination (weighted sum) of the $\mathbf{B}$'s p row vectors

$$\mathbf{c}^{(1)} = \mathbf{a}^{(1)} @ \mathbf{B} = \underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,D} \end{bmatrix}}_{\mathbf{a}^{(1)}} \underbrace{\begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(D)} \end{bmatrix}}_{\mathbf{B}} = a_{1,1} \mathbf{b}^{(1)} + a_{1,2} \mathbf{b}^{(2)} + \dots + a_{1,D} \mathbf{b}^{(D)}$$

We call this the fourth perspective on matrix multiplication – the row-vector-linear-combination perspective.

Figure 2. $\mathbf{B}$'s Linear Combination of Row Vectors

Transpose

Let's switch perspectives again: take Figure 1's matrix multiplication $\mathbf{c}^{(1)} = \mathbf{a}^{(1)}\mathbf{B}$ and transpose it; we get Figure 3, corresponding to

$$\mathbf{c}^{(1)\top} = \mathbf{B}^\top @ \mathbf{a}^{(1)\top}$$

This brings us back to the linear combination of column vectors. The figure below also confirms, from another angle, that the transpose of the matrix multiplication $\mathbf{AB} = \mathbf{C}$ is $\mathbf{B}^\top \mathbf{A}^\top = \mathbf{C}^\top$.

⚠ $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$, and please don't forget to reverse the order.

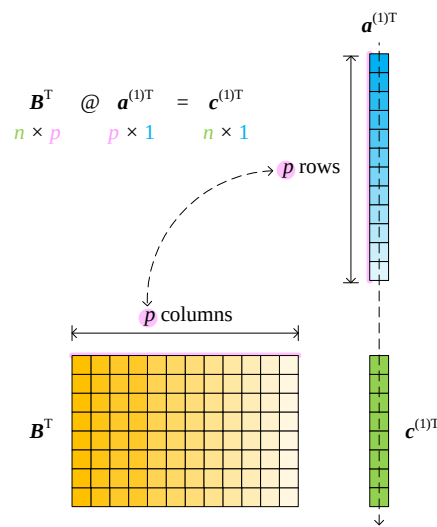


Figure 3. Matrix Multiplication $\mathbf{a}^{(1)}\mathbf{B} = \mathbf{c}^{(1)}$, Transposed

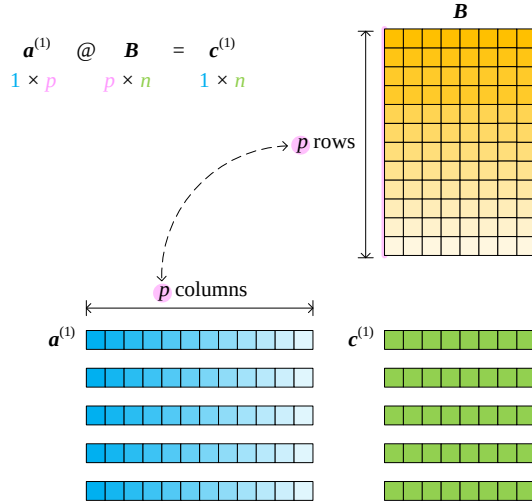
Looking Again at $\mathbf{A} @ \mathbf{B} = \mathbf{C}$

Similarly, we can also use the row-vector-linear-combination perspective to look at the matrix multiplication $\mathbf{C} = \mathbf{A} @ \mathbf{B}$.

As shown in Figure 4, splitting matrix $\mathbf{A}$ into a set of row vectors splits the matrix multiplication $\mathbf{AB} = \mathbf{C}$ into a linear combination of m row vectors!

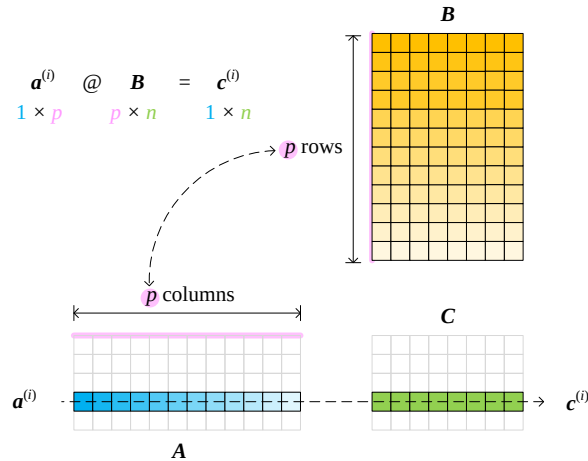
$$\mathbf{C} = \begin{bmatrix} \mathbf{c}^{(1)} \\ \mathbf{c}^{(2)} \\ \vdots \\ \mathbf{c}^{(m)} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(m)} \end{bmatrix} \mathbf{B} = \begin{bmatrix} \mathbf{a}^{(1)}\mathbf{B} \\ \mathbf{a}^{(2)}\mathbf{B} \\ \vdots \\ \mathbf{a}^{(m)}\mathbf{B} \end{bmatrix}$$

Figure 4 tells us that matrix $\mathbf{A}$ has m rows, so the result of the matrix multiplication $\mathbf{A} @ \mathbf{B}$ also has m rows.

Figure 4. Write Matrices $\mathbf{C}$ and $\mathbf{A}$ Respectively as Sets of Row Vectors

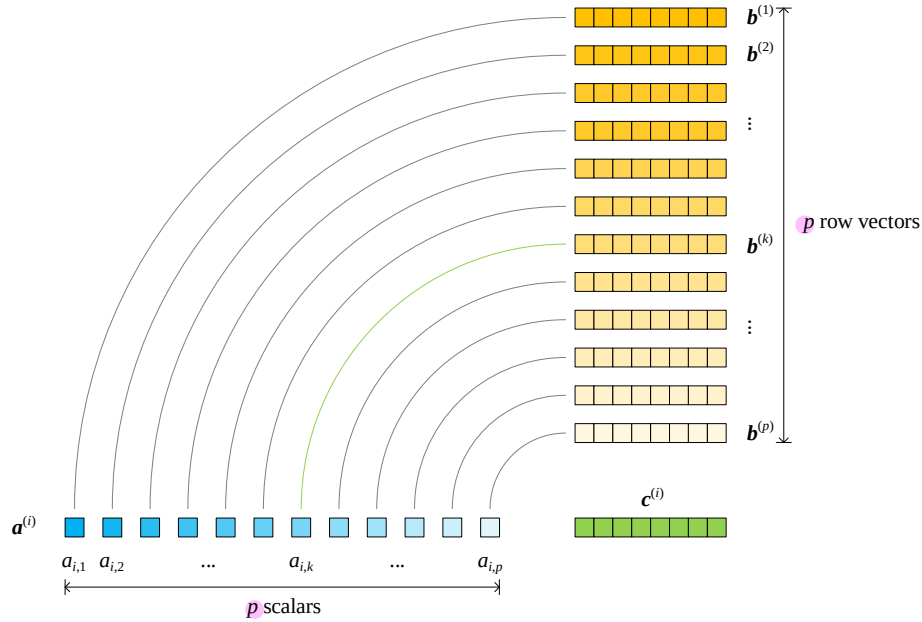
As shown in Figure 5, any row of matrix $\mathbf{C}$ can be computed via the formula below

$$\mathbf{c}^{(i)} = \mathbf{a}^{(i)} @ \mathbf{B}$$

Figure 5. $\mathbf{c}^{(i)} = \mathbf{a}^{(i)} @ \mathbf{B}$ – the Corresponding Matrix Multiplication

Therefore, as shown in Figure 6, matrix $\mathbf{C}$'s i -th row vector $\mathbf{c}^{(i)}$ can be further written as a linear combination of the row vectors of matrix $\mathbf{B}$, with the combination coefficients provided by matrix $\mathbf{A}$'s i -th row vector $\mathbf{a}^{(i)}$'s components, that is

$$\mathbf{c}^{(i)} = \mathbf{a}^{(i)} @ \mathbf{B} = \underbrace{\begin{bmatrix} a_{i,1} & a_{i,2} & \cdots & a_{i,D} \end{bmatrix}}_{\mathbf{a}^{(i)}} \underbrace{\begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(D)} \end{bmatrix}}_{\mathbf{B}} = a_{i,1}\mathbf{b}^{(1)} + a_{i,2}\mathbf{b}^{(2)} + \cdots + a_{i,D}\mathbf{b}^{(D)}$$

Figure 6. Matrix $\mathbf{B}$'s Row-Vector-Linear-Combination Perspective on $\mathbf{c}^j = \mathbf{a}^j \mathbf{B}$

Conversely, after separately computing $\mathbf{a}^{(1)}\mathbf{B}$, $\mathbf{a}^{(2)}\mathbf{B}$, $\mathbf{a}^{(3)}\mathbf{B}$, and so on, stacking them in order, one below another, gives matrix $\mathbf{C}$, that is

$$\begin{bmatrix} \mathbf{a}^{(1)}\mathbf{B} \\ \mathbf{a}^{(2)}\mathbf{B} \\ \vdots \\ \mathbf{a}^{(m)}\mathbf{B} \end{bmatrix} = \begin{bmatrix} \mathbf{c}^{(1)} \\ \mathbf{c}^{(2)} \\ \vdots \\ \mathbf{c}^{(m)} \end{bmatrix}$$

The First Example

Let's look at the matrix-multiplication example covered earlier in this book; using the row-vector-linear-combination method, compute $\mathbf{c}^{(1)}$

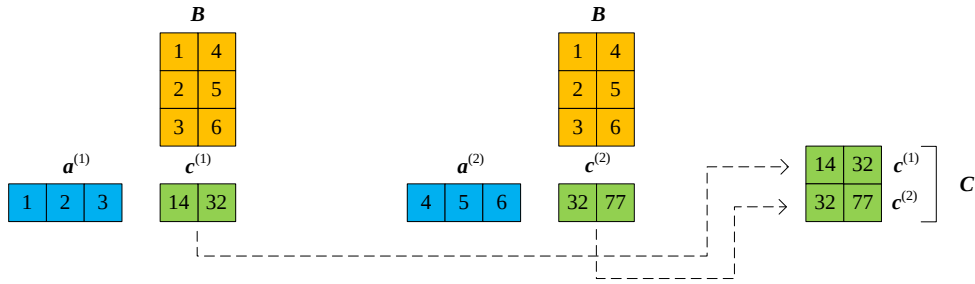
$$\mathbf{c}^{(1)} = \mathbf{a}^{(1)} @ \mathbf{B} = [1 \quad 2 \quad 3] @ \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = 1 \times [1 \quad 4] + 2 \times [2 \quad 5] + 3 \times [3 \quad 6] = [14 \quad 32]$$

First compute $\mathbf{c}^{(2)}$

$$\mathbf{c}^{(2)} = \mathbf{a}^{(2)} @ \mathbf{B} = [4 \quad 5 \quad 6] @ \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = 4 \times [1 \quad 4] + 5 \times [2 \quad 5] + 6 \times [3 \quad 6] = [32 \quad 77]$$

As shown in Figure 7, then arrange the row vectors $\mathbf{c}^{(1)}$ and $\mathbf{c}^{(2)}$ in order to get matrix $\mathbf{C}$

$$\mathbf{C} = \mathbf{A} @ \mathbf{B} = \begin{bmatrix} \mathbf{c}^{(1)} \\ \mathbf{c}^{(2)} \end{bmatrix} = \begin{bmatrix} [14 \quad 32] \\ [32 \quad 77] \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

Figure 7. Arranging Row Vectors $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ in Order to Get Matrix $\mathbf{C}$

The Second Example

Using the row-vector-linear-combination method, compute $\mathbf{d}^{(1)}$

$$\mathbf{d}^{(1)} = \mathbf{b}^{(1)} @ \mathbf{A} = [1 \ 4] @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = 1 \times [1 \ 2 \ 3] + 4 \times [4 \ 5 \ 6] = [17 \ 22 \ 27]$$

First compute $\mathbf{d}^{(2)}$

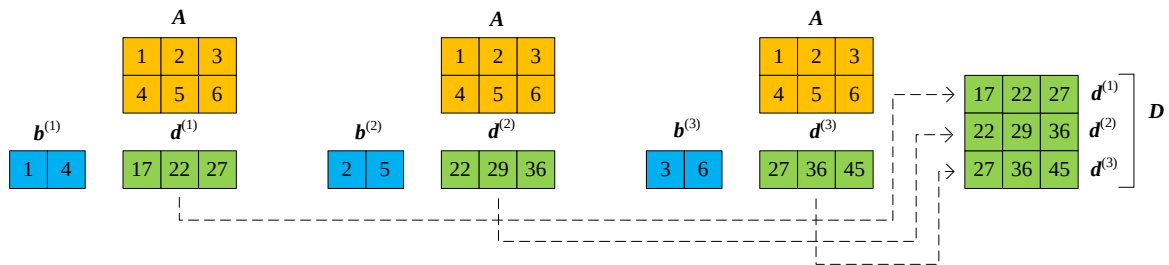
$$\mathbf{d}^{(2)} = \mathbf{b}^{(2)} @ \mathbf{A} = [2 \ 5] @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = 2 \times [1 \ 2 \ 3] + 5 \times [4 \ 5 \ 6] = [22 \ 29 \ 36]$$

Finally compute $\mathbf{d}^{(3)}$

$$\mathbf{d}^{(3)} = \mathbf{b}^{(3)} @ \mathbf{A} = [3 \ 6] @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = 3 \times [1 \ 2 \ 3] + 6 \times [4 \ 5 \ 6] = [27 \ 36 \ 45]$$

As shown in Figure 8, arrange the row vectors $\mathbf{d}^{(1)}$, $\mathbf{d}^{(2)}$, and $\mathbf{d}^{(3)}$ in order to get matrix $\mathbf{D}$

$$\mathbf{D} = \mathbf{B} @ \mathbf{A} = \begin{bmatrix} \mathbf{d}^{(1)} \\ \mathbf{d}^{(2)} \\ \mathbf{d}^{(3)} \end{bmatrix} = \begin{bmatrix} [17 \ 22 \ 27] \\ [22 \ 29 \ 36] \\ [27 \ 36 \ 45] \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

Figure 8. Arranging Row Vectors $\mathbf{d}^{(1)}$, $\mathbf{d}^{(2)}$, and $\mathbf{d}^{(3)}$ in Order to Get Matrix $\mathbf{D}$ 

LA_03_04_01.ipynb completes the fourth-perspective matrix-multiplication computations above – please study it on your own.

Visualizing the Fourth Perspective on Matrix Multiplication with a Heatmap

Next, let's use a heatmap to visualize the fourth perspective on matrix multiplication.

Figure 9 uses a heatmap to show how to compute $\mathbf{a}^{(1)} @ \mathbf{B}$.

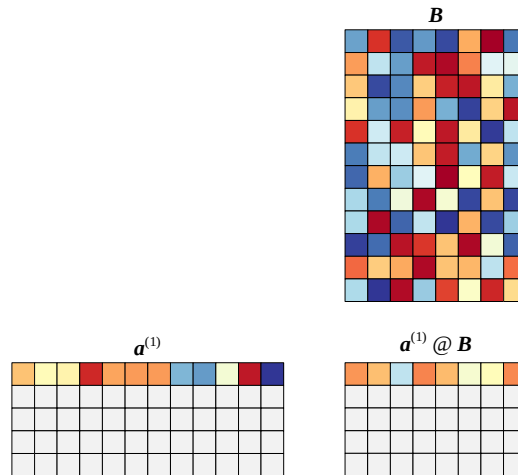


Figure 9. Heatmap Showing $\mathbf{a}^{(1)} @ \mathbf{B}$

Since $\mathbf{A}$ has 5 rows, there are matrix multiplications like $\mathbf{a}^{(1)} @ \mathbf{B}$, 5 in total; the results are shown in Figure 10. Stacking their results in order, top to bottom, gives matrix $\mathbf{C} = \mathbf{A} @ \mathbf{B}$.

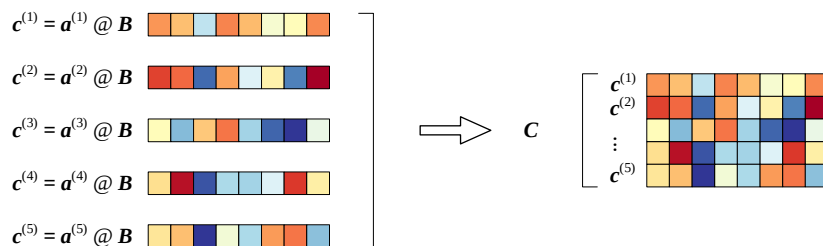


Figure 10. 5 $\mathbf{a}^{(i)} @ \mathbf{B}$ Stacked Top to Bottom Give Matrix $\mathbf{C} = \mathbf{A} @ \mathbf{B}$



LA_03_04_02.ipynb uses `seaborn.heatmap()` to draw the heatmap above, visualizing the fourth perspective on matrix multiplication – please study it on your own.

Code 1 uses a custom function to complete the matrix multiplication using the row-vector-linear-combination perspective. The code is similar to the “column-vector-linear-combination perspective” from the previous section — please compare and analyze this code alongside that one.

Code 1. A Custom Python Function to Compute Matrix Multiplication, the Row-Vector-Linear-Combination Perspective |  LA_03_04_03.ipynb

```
## 初始化
import numpy as np

## 定义行向量线性组合函数
def row_LC(B, coefficients):
    p, n = B.shape
    row_LC_result = np.zeros(n)
    # 初始化行向量组合结果

    for k in range(p):
        row_LC_result += coefficients[k] * B[k, :]
        # 每行乘系数并相加
    return row_LC_result

## 定义矩阵乘法函数（行线性组合法）

def matrix_multiplication_row_LC(A, B):
    # 获取矩阵 A 和 B 的形状
    m, p_A = A.shape
    p_B, n = B.shape

    # 检测矩阵形状是否符合矩阵乘法规则
    if p_A != p_B:
        raise ValueError('Dimensions do not match')

    # 初始化结果矩阵 C，形状 (m, n)，初始值设为 0
    C = np.zeros((m, n))

    for i in range(m):
        coeffs = A[i, :]
        # 取出 A 的第 i 列，作为线性组合的系数
        C[i, :] = row_LC(B, coeffs)
        # 计算 B 行向量的线性组合

    return C

## 矩阵乘法
A = np.array([[1, 2, 3],
              [4, 5, 6]])
B = A.T

## 矩阵乘法
matrix_multiplication_row_LC(A, B)
matrix_multiplication_row_LC(B, A)
```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Use the row-vector-linear-combination method to compute the following pairs of matrix multiplications

► $\mathbf{a}^{(1)} = [1 \ 2], \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

► $\mathbf{a}^{(1)} = [1 \ 2 \ 3], \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

► $\mathbf{a}^{(1)} = [1 \ 2], \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

► $\mathbf{a}^{(1)} = [1 \ 2 \ 3], \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

Q2. Use the fourth perspective on matrix multiplication to expand the following pairs of matrix multiplications

► $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

► $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

► $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad \mathbf{B} = \mathbf{A}^T$

► $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad \mathbf{B} = \mathbf{A}^T$

Q3. Combine the following two matrix multiplications into one.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} @ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} @ \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Q4. Combine the following three matrix multiplications into one, and give the result.

$$[1 \ 0 \ 0] @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad [0 \ 1 \ 0] @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad [0 \ 0 \ 1] @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Q5. Combine the following three matrix multiplications into one, and give the result.

$$[0 \ 0 \ 1]@ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad [1 \ 0 \ 0]@ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad [0 \ 1 \ 0]@ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Q6. Compare the results of Q4 and Q5, and explain the similarities and differences between them.